

[illegible]

Fig. 5.6: Session Log View – Timestamped Sensor Data Log (2026-04-09)

Fault Detection and Classification

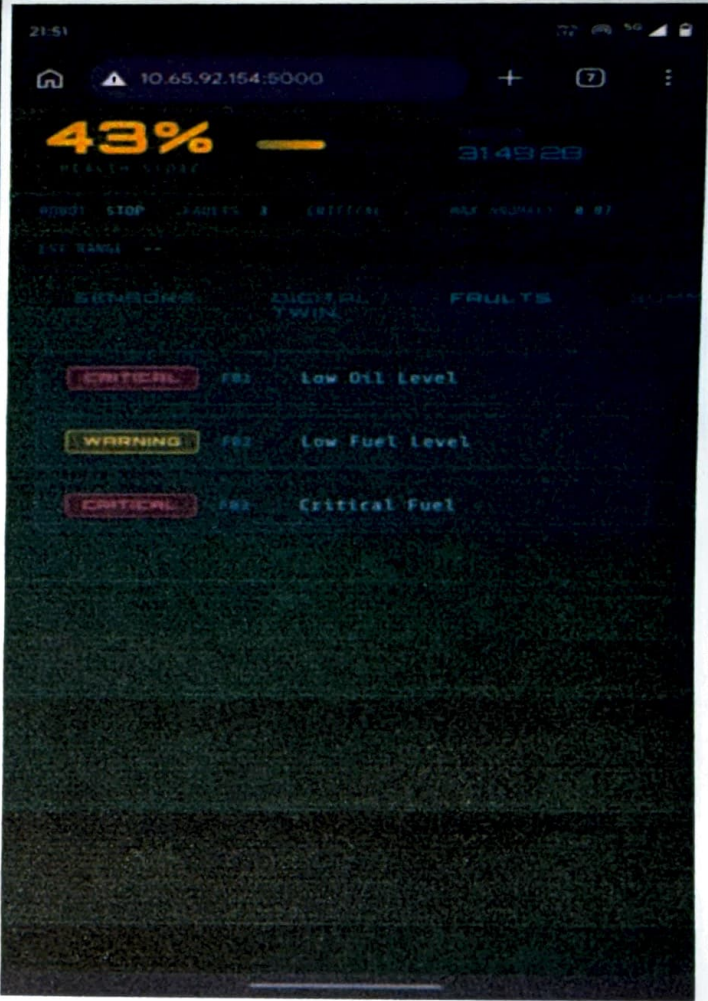
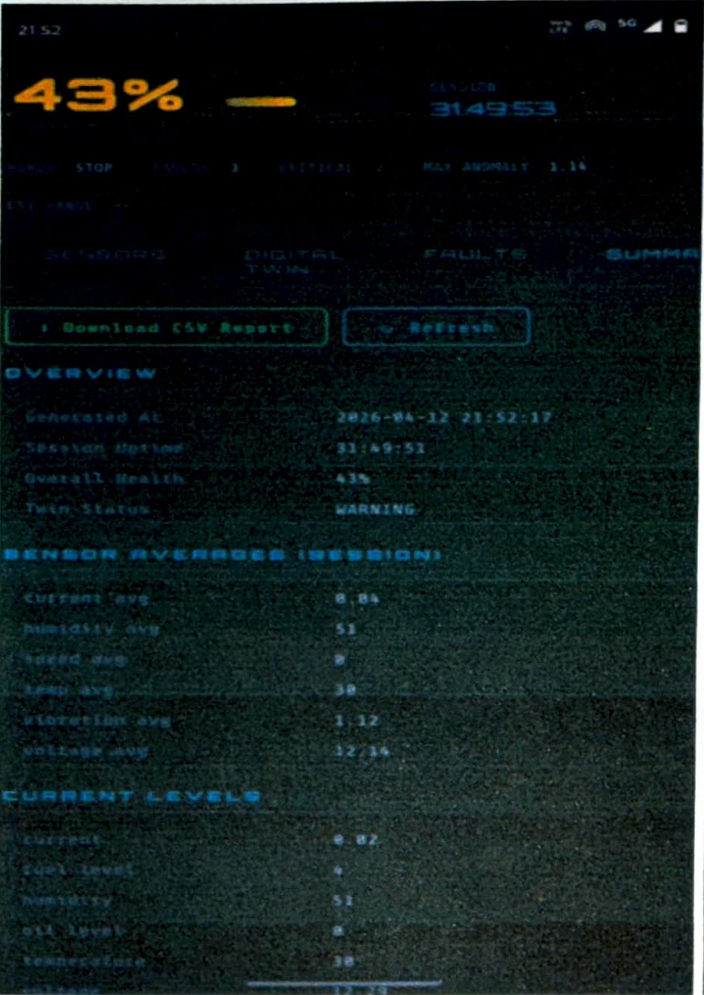


Fig. 5.4: Faults Tab – Active Fault List (Critical: Low Oil, Critical Fuel; Warning: Low Fuel) Fig. 5.5: Summary Tab – Session Overview and Sensor Averages

Digital Twin Component Health Status and Anomaly Detection

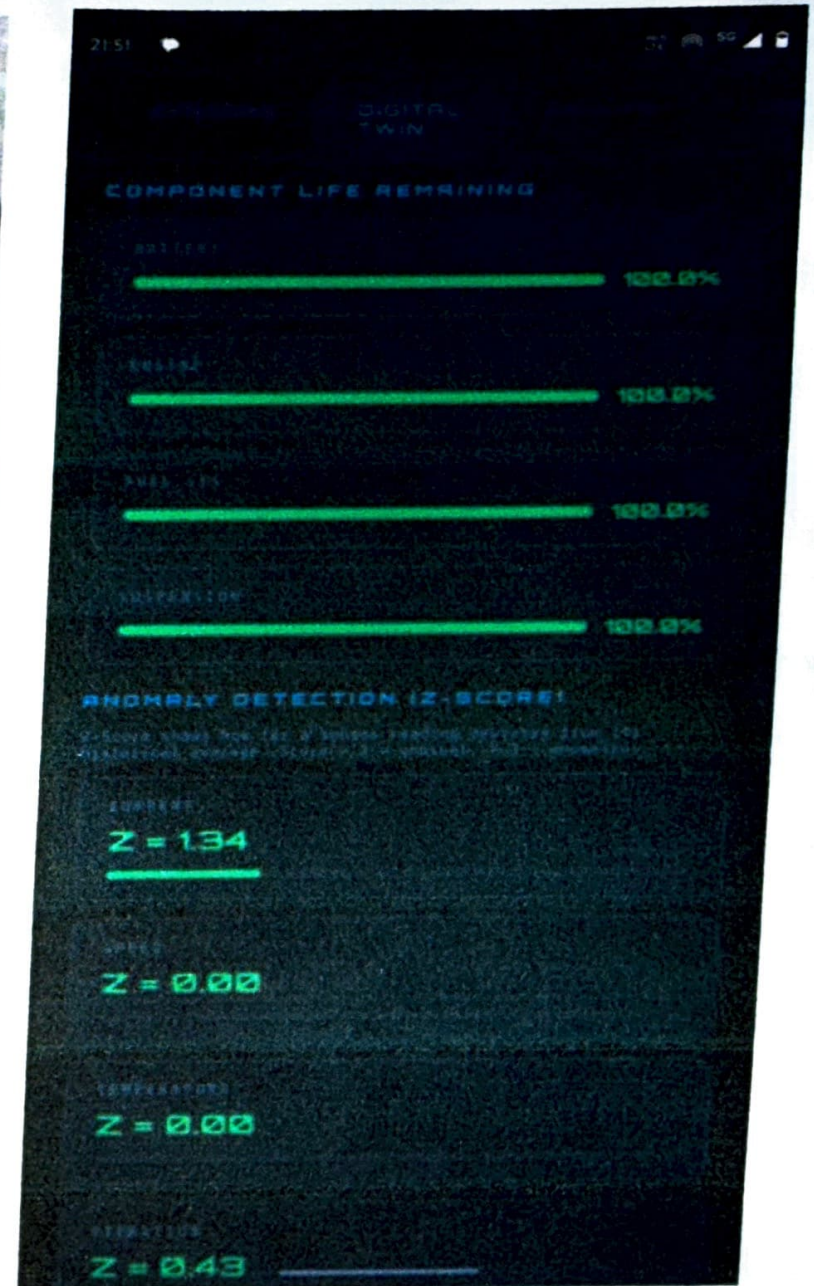
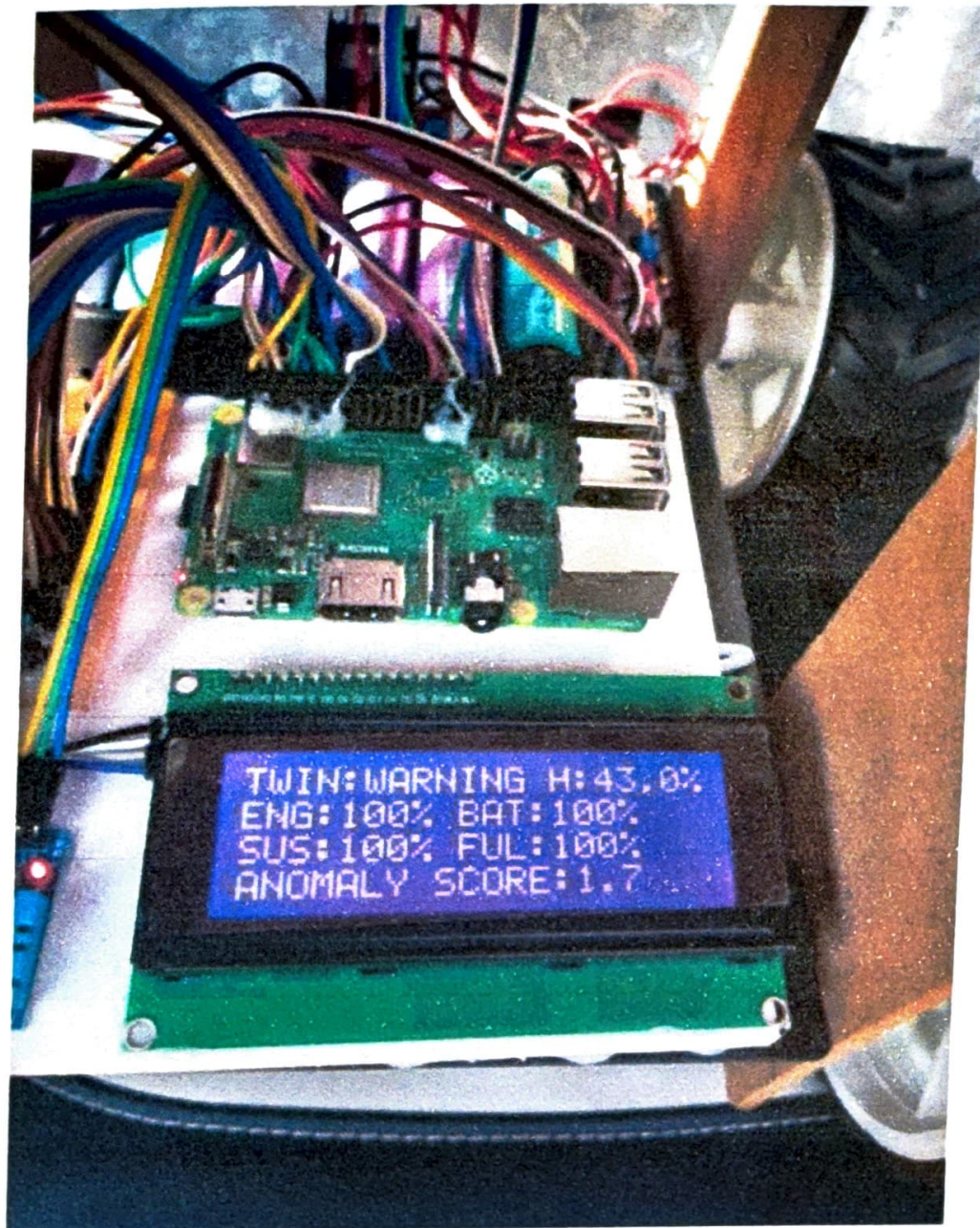


Fig. 5.3: Digital Twin Tab – Component Life and Anomaly Detection (Z-Score)

Sensor Data Acquisition

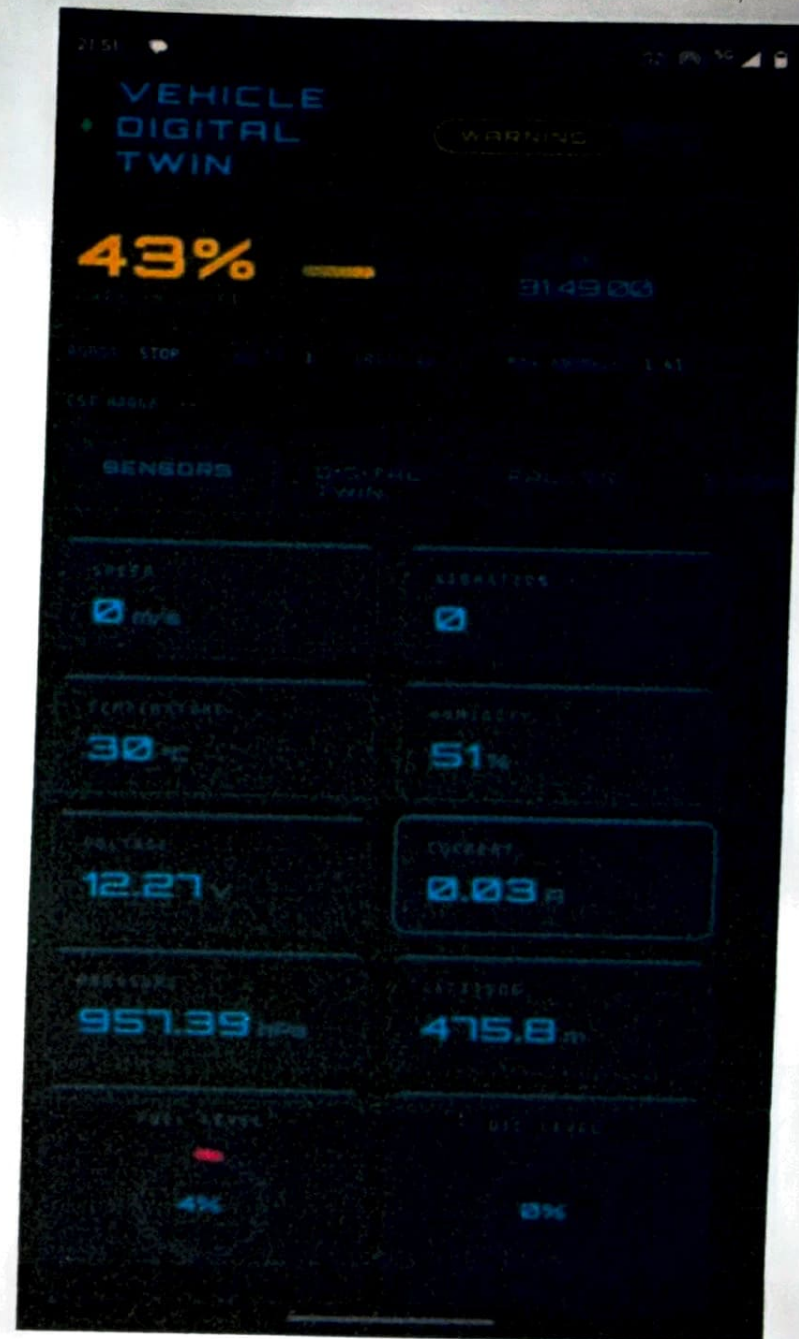
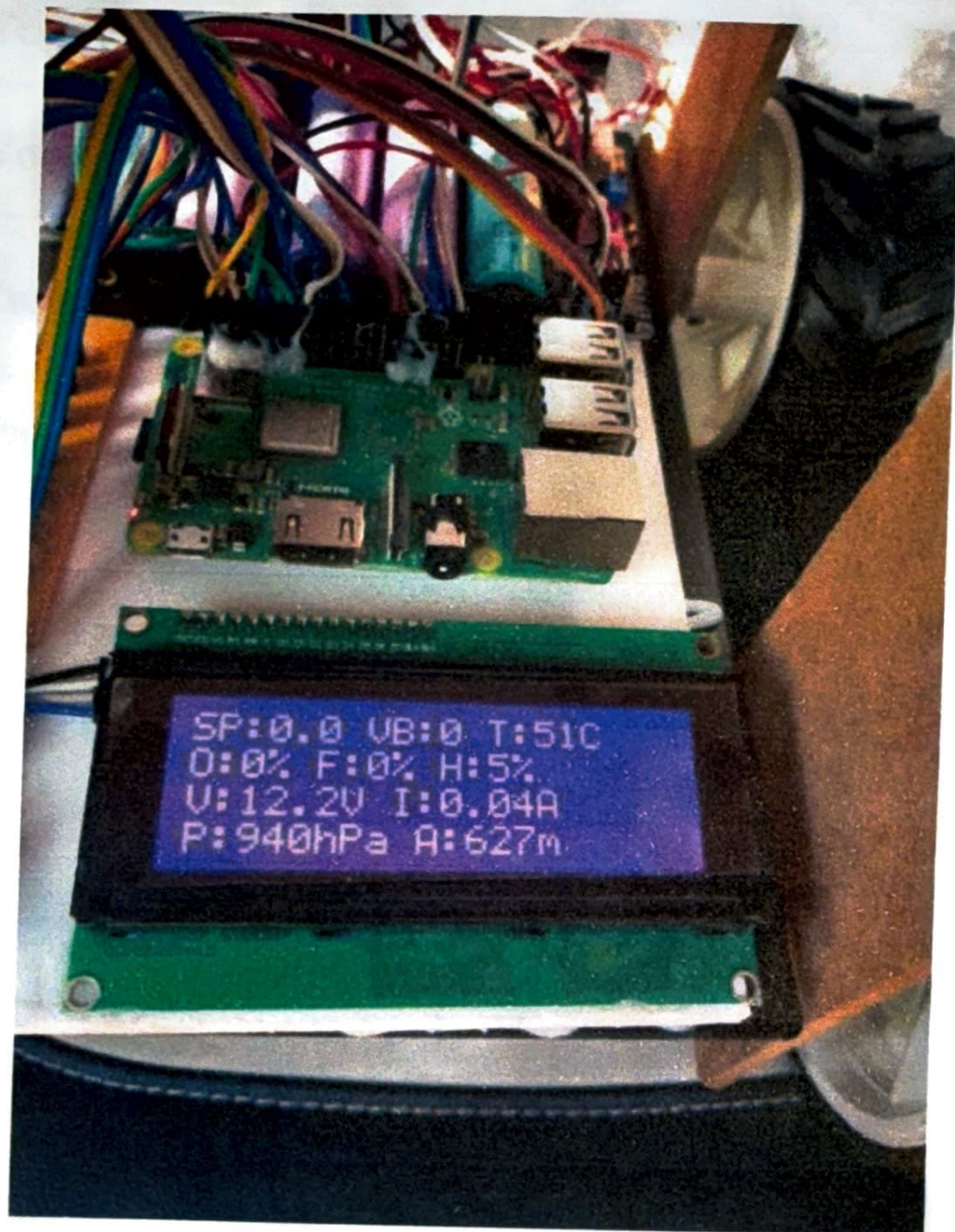


Fig. 5.2: Sensors Tab – Real-Time Vehicle Dashboard (43% Health Score)

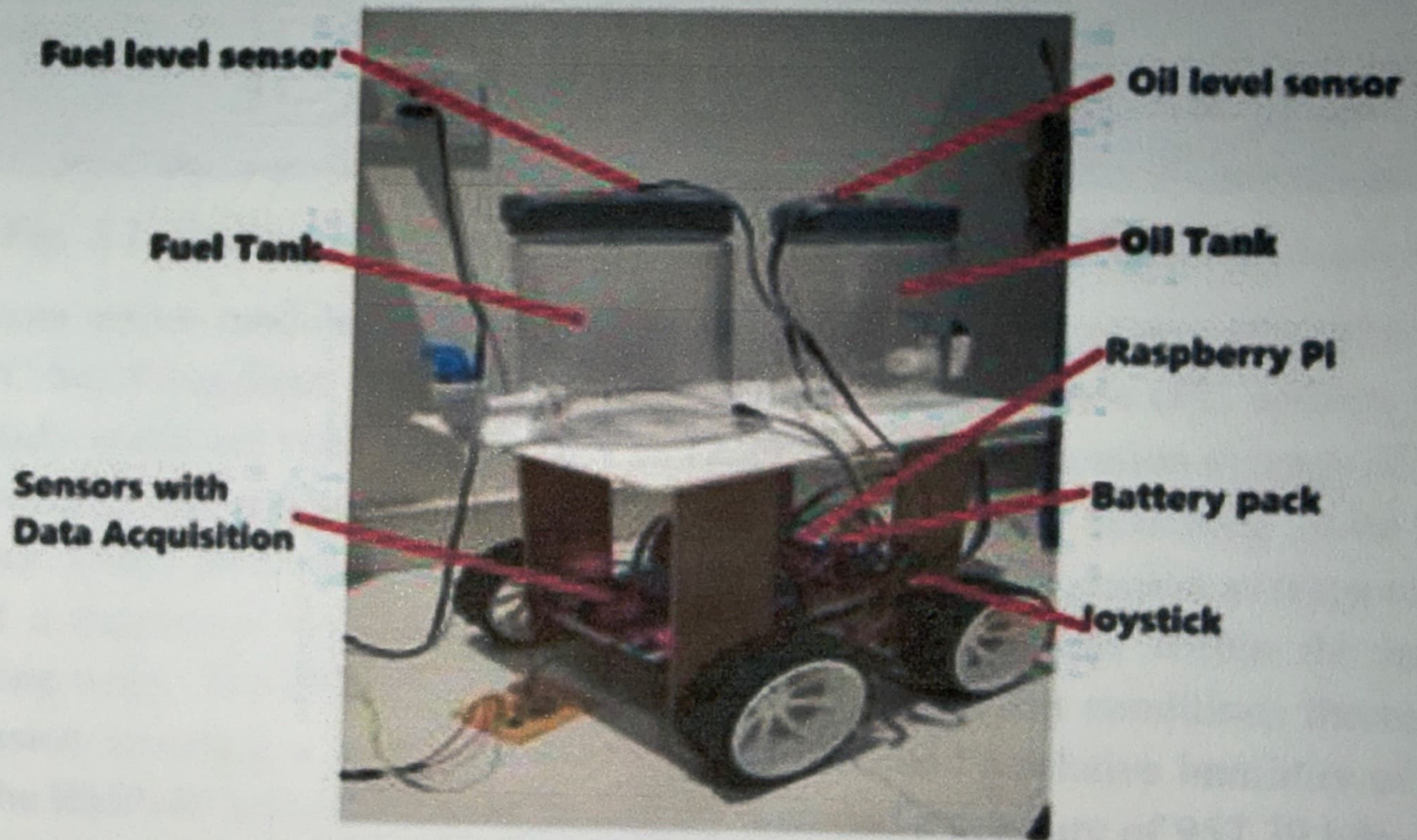


Fig. 5.1: Final Hardware Prototype of the Digital Twin Vehicle Health Monitoring System